

# How Much Local Data Is Enough?

---

A symbolic gate was built to answer that, it was **believed**, it was **falsified across all 380 site combinations**, and the criteria it read were re-taxonimized into a different set of gatekeepers. This is the arc in ten slides.

**Rodrigo Huber Marques Moreira Mendes** · Fernanda Baião · Reinaldo Castro Souza

PUC-Rio, Industrial Engineering | 20 Brazilian MERRA-2 sites, 2017–2020 | fast talk, 10 slides

# One number decides whether transfer is worth it

$n^*$  — the data budget: the local days after which fine-tuning *permanently* beats the zero-shot source model

## The setup

Source **Galeão / GIG** (Rio de Janeiro airport). Random Forest (200 trees, depth 10, seed 42) on MERRA-2 daily means — TA, PR, NV, TOA → GHI, 2017–2020. **20 sites**, so  **$20 \times 19 = 380$  ordered transfer combinations**.

## The measurement

Few-shot = a **residual corrector** trained on K local days on top of the zero-shot prediction.  $n^* = \min\{K : n\text{RMSE\_FTL}(K') < n\text{RMSE\_ZTL} \text{ for all } K' \geq K\}$ . No such K →  $n^* = \infty$  (right-censored).

**WHY PERMANENCE** The strict rule matters: several findings that look significant under a lenient “FTL wins somewhere” rule **collapse or reverse** under permanence. That distinction is part of the honesty record, not a technicality.

Roadmap: **I.** the question → **II.** the gate as designed and as believed → **III.** falsification across all 380 combinations → **IV.** the new gatekeepers → **V.** what survives.

# The Neurosymbolic Semantic Gate: six epistemic faculties

Each layer reads a measurable property of a source→target pair and emits PASS / SOFT / WARN / FAIL

| Layer | Faculty (PRE) | Test / threshold, original calibration                                                  | Weight |
|-------|---------------|-----------------------------------------------------------------------------------------|--------|
| L1    | Semantic      | TOA irradiance ratio; PASS if $0.998 \leq r \leq 1.073$                                 | 0.15   |
| L2    | Operational   | Two-sample KS on de-seasonalised cloud cover; PASS if $p > 0.05$                        | 0.15   |
| L3    | Mathematical  | Pearson r on 12-month GHI climatology; PASS $\geq 0.90$ , WARN 0.55–0.90, FAIL $< 0.55$ | 0.25   |
| L4    | Causal        | Whittaker distance in (T, log P) space; PASS $< 1.50$ , WARN $< 2.42$ , else WARN/FAIL  | 0.25   |
| L5    | Explanatory   | Holdridge life-zone <i>and</i> Köppen class identity conjunction                        | 0.10   |
| L6    | Metatheoretic | Reflexive self-audit of the verdict against empirical $n^*$                             | 0.10   |

## Aggregation

Horn clause  $P1 \wedge P2 \wedge (P3 \neq \text{FAIL}) \wedge (P4 \neq \text{FAIL}) \wedge P5 \rightarrow$  **APPROVED / WARNING / REJECTED**; scalar  $\Omega = \sum w_k s_k$  as a convenience.

## H1, pre-registered BELIEVED

**Gate-APPROVED pairs reach their crossover  $n^*$  earlier than gate-REJECTED pairs.** Climatological similarity  $\Rightarrow$  transferable parameters  $\Rightarrow$  less local data needed.

# It looked supported — and the caveats were already in the file

Historical statistics only. Do not cite these as current results.

| Statistic (era)             | Value  |
|-----------------------------|--------|
| Mann–Whitney U, one-sided   | 6.5    |
| p-value                     | 0.0075 |
| Rank-biserial r             | −0.684 |
| WARNING-tier LOOCV accuracy | 94.7%  |

Groups: APPROVED  $n = 10$  vs. REJECTED  $n = 2$ . WARNING: 7 of the 19 single-source targets,  $n^*$  spanning 1–180 days.

Qualifications recorded at the time

- ▶ REJECTED had  $n = 2$ : one reordering of  $n^*$  collapses significance.
- ▶ 94.7% was LOOCV from a **single source** (GIG) across 19 targets — flagged exploratory.
- ▶ The WARNING band spanned **1–180 days**; the proposed L4 tiebreaker was never validated.
- ▶ L6's self-audit already flagged Viçosa — gate WARNING, the *best* observed partner — as a conservative error.

**The tell.** Every one of those qualifications sat in the code and the specification. What failed was not detection — it was the **weight** they were given in the framing.

# All 380 combinations, leave-one-source-out: an *exact* null

Same test, honest protocol, strict permanence  $n^*$  —  $H_1$  is falsified

## H1 FALSIFIED $p = 1.0$

Mann–Whitney  $U = 27.0$ ,  $p = 1.0$  over all  $20 \times 19 = 380$  directed pairs. At the practical operating point (9 APPROVED vs. 6 REJECTED single-source targets)  $U = 27.0$  lands on its null expectation  $\mu = 27$  exactly:  $Z = 0.0000$ , effect size  $r = 0.0000$ . The rank distributions overlap *perfectly*. And  $p = 1.0$  means APPROVED pairs had the **larger** median  $n^*$  — the direction opposite to  $H_1$ .

## The counter-example that ends it

**GIG** → **São Domingos**, rated **REJECTED**, crosses at  $n^* = 46$  d — faster than *every* APPROVED Rio-area pair (best: GIG → Barra da Tijuca, 50 d). **GIG** → **Marambaia**, rated **APPROVED**, is the slowest of all 19 at  $n^* = 154$  d.

## What broke between the two readings

- ▶  $n = 12$  calibration set → **380 pairs**;
- ▶ single-source LOOCV → **leave-one-source-out**;
- ▶ lenient crossover → **strict permanence rule**.

No new data. Only an honest protocol.

# “You just mis-set the thresholds” — foreclosed by search

A confusion matrix at the real operating point, then the entire threshold grid

| Gate verdict as a classifier of fast transfer ( $n^* \leq 90$ d), $n = 19$ | Value                 | Threshold-robustness sweep over L1 / L2 / L4 and the WARNING tier | Value |
|----------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------|-------|
| True positives / false positives                                           | 6 / 3                 | Valid threshold combinations tested                               | 1035  |
| Precision · recall · specificity                                           | 0.667 · 0.667 · 0.700 | Combinations with one-sided $p < 0.05$                            | 0     |
| Balanced accuracy · F1                                                     | 0.683 · 0.667         | Surviving BH-FDR correction ( $q < 0.05$ )                        | 0     |
| Matthews correlation coefficient                                           | 0.367                 | Best one-sided $p$ found (effect $r = -0.058$ )                   | 0.417 |
| Fisher exact test on the $2 \times 2$ table                                | $p = 0.179$ (n.s.)    | Grid-aware permutation $p$ , 2000 shuffles                        | 0.938 |

**READ THE SIGN** The single best result the whole 1035-combination search could find has the **wrong sign** — it favours REJECTED sites crossing faster. Against a Westfall–Young-style null built by re-searching the entire grid on 2000 shuffles, the real grid's best result is **indistinguishable from noise**. The failure is not a calibration artefact.

# The built-in control: transfer Galeão onto itself

Similarity pinned at its theoretical maximum on every layer — and the gate is still silent

| Layer         | Quantity                          | Value                  |
|---------------|-----------------------------------|------------------------|
| L1            | TOA / reference ratio             | 1.0                    |
| L2            | KS p on cloud-cover STL residuals | 1.0                    |
| L3            | Köppen class                      | Aw vs. Aw              |
| L4            | Whittaker T–P distance            | 0.0                    |
| L5            | Holdridge life zone               | identical              |
| L6            | IBGE biome (post-hoc)             | identical              |
| Gate decision |                                   | APPROVED, all six PASS |

**The clean statement.** The most-similar pair is not the fastest, and the fastest pair is not the most similar.

**And yet it is not the fastest**

Self-transfer crosses at **n\* = 22 d** (FTL beats ZTL on 160/180 windows) — **2nd of 20**. The one pair that beats it, **Nova Brasília Urucuí at n\* = 15 d**, is a climatically distant Cerrado site the gate **REJECTS**.

**What actually produced the fast crossover**

**Temporal recency**, not similarity. Fine-tuning on the most recent three weeks recovers non-stationary signal a forest trained on 1071 days averages away. **No static similarity layer encodes recency.**

# It was never one hypothesis — the whole similarity family falls

Stage-0 ledger: five candidate mechanisms, refutation criteria fixed before each test

| Candidate             | Claim about what governs n*                          | Outcome                                                                                                                                                                                  | Status  |
|-----------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| H1 gate verdict       | APPROVED pairs cross earlier than REJECTED           | U = 27.0, p = 1.0 over 380 pairs                                                                                                                                                         | REFUTED |
| H1' DTW / two regimes | Smaller climatological distance ⇒ smaller n*         | Permanence rule p = 0.671, median SIMILAR 59 d < DISTANT 119 d (opposite); mixed-effects coef. +0.009, p = 0.058, wrong sign. Lenient-grid p = 0.017 was a crossover-definition artefact | REFUTED |
| H1" CORAL / SA        | Distributional alignment shrinks ZTL error, hence n* | SA worse than plain ZTL at <b>19/19</b> sites (median nRMSE 0.166 vs. 0.157); CORAL median 0.159 > ZTL despite 11/19 site-level wins                                                     | REFUTED |
| H0 noise floor        | No label-free variable predicts n*                   | Bayesian variance decomposition: 9% source + 16% target explained, <b>75% residual</b> ; rolling-origin CV median 208-day across-fold range                                              | LIVE    |

**WHY THE PATTERN IS DIAGNOSTIC** CORAL and SA are **second-moment** aligners — they whiten, re-colour and rotate to match covariance. If the dominant ZTL error is a **first-moment** offset, no amount of covariance alignment can touch it. That is exactly the failure observed, and it is what abducted H2. Abduction suspects; it does not confirm.

# The same six criteria, re-taxonimized

The layers stop being named for the kind of knowledge they externalize, and start being named for what they read

| L  | PRE — epistemic faculty          | POST — gatekeeper   | POST test                                                             |
|----|----------------------------------|---------------------|-----------------------------------------------------------------------|
| L1 | Semantic                         | <b>Astronomical</b> | TOA ratio $\in [0.85, 1.15]$ — widened from $[0.998, 1.073]$          |
| L2 | Operational                      | <b>Atmospheric</b>  | KS test on STL-deseasonalised cloud cover (NV)                        |
| L3 | Mathematical (seasonal r)        | <b>Köppen</b>       | Köppen class identity — PASS / SOFT / REJECT                          |
| L4 | Causal                           | <b>Whittaker</b>    | Scaled Euclidean distance in (T, P) space                             |
| L5 | Explanatory (Holdridge + Köppen) | <b>Holdridge</b>    | Life-zone identity — PASS / WARN, no hard REJECT                      |
| L6 | Metatheoretic (self-audit)       | <b>IBGE Biome</b>   | <b>Post-hoc only, casts no vote — IBGE shapefile point-in-polygon</b> |

## The consequential change: L6 is disenfranchised

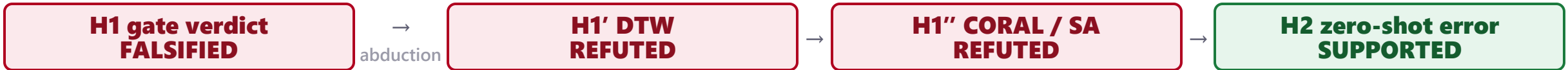
The faculty that audited the gate against its own record becomes a physical criterion **excluded from the Horn-clause vote** (`nsg:isPostHoc true`) — diagnostic only.

## And L5 splits in two

What PRE bundled as one conjunction becomes **two independent criteria**.  
Ontology closure grows **1,375 → 1,943 triples**, all checks PASS.

# The real gatekeeper was a quantity the gate already read

H2: zero-shot baseline error schedules the data budget — abducted, then confirmed five ways



| Confirmatory test for H2         | Independent axis                  | Result                                                               |
|----------------------------------|-----------------------------------|----------------------------------------------------------------------|
| Single-source Spearman / Pearson | Small sample, one source site     | $p = -0.57 \text{ to } -0.71, p \leq 0.013$                          |
| Crossed mixed-effects, 380 pairs | Pseudoreplication corrected       | $\text{coef. } -8.45, p = 1.1 \times 10^{-5}$                        |
| Within-source Wilcoxon           | Non-parametric                    | 18/20 sources, $p = 0.001$                                           |
| Cox proportional hazards         | Handles the never-crossover pairs | $\text{HR} = 1.57/\text{SD}, p < 10^{-4}, \text{LOSO C-index } 0.66$ |
| Hierarchical Bayesian            | Different inferential paradigm    | $b = -0.47/\text{SD}, 94\% \text{ HDI } [-0.61, -0.31]$              |

What the era's numbers became

WARNING tier **94.7% LOOCV** → **63.7–73.9% LOSO** (n = 380, permutation p = 0.0001). Budget regressor: GBT, **LOSO MAE = 23.1 d** on 227 pairs.

The bound that stays on

H0 was never eliminated: **75% of the variance in n\* is residual**. Five tests, same 380 pairs — multi-method, **single-dataset**.

**The contribution is the arc.** Three candidates eliminated by pre-specified criteria, a fourth left standing as an honest bound, one confirmed — and a formal vocabulary that outlived the predictive claim built on it.